

Simple Transformer with Single Leaky Neuron for Event Vision

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Abstract

There is an increasing interest in integrating the self-attention mechanism and transformer architecture in event-based vision. Several event-based and spiking neural network transformers have been proposed with complex architectures. We propose a simple transformer with a multi-head attention mechanism, a feature extractor, and a single leaky neuron. We have used ResNet, an effective feature extractor for RGB images, on frame-based event data. Experimental results show that our architecture surpasses many event and spiking-based methods, including transformers, and achieves competitive performance in DVS Gesture, N-MNIST, and CIFAR10-DVS datasets. Notably, we achieve an accuracy of 98.3% on the DVS Gesture and 99.3% on the N-MNIST dataset. Source code available at [GitHub](#).

1. Introduction

From the first-generation multi-layer perceptron (MLP) to numerous cutting-edge methods in the second-generation deep neural networks (DNNs), artificial neural networks (ANNs) have grown in capability in the last decade. ANNs have achieved excellent results in computer vision, natural language processing, speech, autonomous driving, and many other applications. ANNs [25] require significant annotated data, broad availability of high-performance computing, and powerful Graphics Processing Units (GPUs).

However, ANNs still fall short of biological neural networks regarding learning capabilities and energy efficiency. Despite being inspired by the brain, ANNs differ in structure, neural computations, and learning rules. ANNs have no memory and each of their neuron uses a continuously differentiable activation function. Spiking neural networks (SNNs), sometimes known as the third generation [14] of neural networks, result from this observation and can potentially overcome ANN bottlenecks.

Spiking neurons interact with each other using discrete electrical signals known as spikes, and they work in continuous time. SNNs can be used on specially designed

neuromorphic hardware, which results in much less energy consumption. Since they are similar to biological neurons, SNNs work well with sparsity and are highly compatible with temporal code [11]. Even though SNNs currently perform worse than DNNs, the difference is closing on some tasks, and SNNs usually need a lot less energy to function. However, SNNs are still challenging to train overall, primarily because of their intricate neuronal dynamics and non-differentiable spike operations. However, while large pretrained vision models like Vision Transformers and ResNet are excellent at extracting spatial data, they cannot process temporal information.

To bridge this gap, we propose a Simple Transformer, a novel architecture that integrates the spatial strengths of pretrained backbones with the temporal dynamics of spiking neurons. This model leverages pretrained spatial feature extractors and spiking temporal processors to achieve competitive performance on neuromorphic datasets. The main contributions of this paper can be summarized as follows:

- **Pretrained Backbone:** We leverage pretrained ResNet on ImageNet to extract spatial features from frame-based representations of event data. We slightly modified ResNet architecture to accept 2-channel polarity as input.
- **Spiking processor and residual connections:** We added a spiking layer with Leaky Integrate and Fire node to process temporal dynamics across multiple timesteps. To speed up the learning process, we used residual connections with the spiking processor.
- **We propose the Simple transformer architecture** with Multi-head attention layers to capture long-range temporal dependencies.
- **Experiment results** show that our architecture outperforms spiking [5, 6, 21, 24] and event [2, 3, 22] based networks, as well as some transformers [18–20].

2. Related Work

Event-Based Vision Event cameras [8] are revolutionizing the way that visual information is processed. This is

because they sample light according to the scene dynamics rather than using a clock unrelated to the viewed scene. Their benefits include minimal power consumption, a very high dynamic range (140 dB compared to 60 dB for ordinary cameras), and very high temporal resolution and low latency (both in the order of microseconds).

Event data can be represented in mainly two categories: event representation such as graphs [3], group tokens [18], and point clouds [23] which maintain the sparsity of event data. The other representation, called Frame-based representation groups events into dense frame-like arrays, ignoring the event data’s sparsity yet facilitating further learning. TORE volumes [2] a new event representation compactly stores spike information. Recurrent Neural Networks [22] have been utilised to process event information. Recently, Event-based transformers [18, 19] have been proposed which achieve significant performance improvement over event neural network architectures. Frame-based representations serve as the foundation for our work.

Spiking Neural Networks (SNNs) leverage spiking neurons [25], such as the Leaky Integrate-and-Fire (LIF) model, to process temporal data. Recent advances include learnable membrane time constants (e.g., Parametric LIF), which enhance temporal learning. The emerging success of surrogate gradient learning [16] led to the significant development of Spiking Convolutional Neural Networks (SCNNs), which are widely utilized to handle difficult vision tasks including segmentation, object recognition and detection. Neuromorphic Data Augmentation [13] performs geometrical augmentations on event-based data. Spiking Convolutional Recurrent Neural Network [24] architecture preserves the spatial and temporal relations from event-based sequence data by utilizing both convolution operation and recurrent connection. Spiking ResNet [10] incorporate residual structure in a SCNN with ResNet-based architecture. SEW ResNet [5] effectively trains a 152-layer SCNN directly by carefully examining the identity mapping in directly-trained spiking residual networks. Spiking DS-ResNet [7], presents a multi-level firing (MLF) approach based on spiking dormant-suppressed residual network (spiking DS-ResNet) and the existing spatio-temporal back propagation (STBP) method. [6] introduces a training approach which can learn both the synaptic weights and the membrane time constants of SNNs. They demonstrated that adding learnable membrane time constants can accelerate learning and reduce the network’s sensitivity to beginning values.

Spiking Vision Transformers Spikformer [28], first spiking vision transformer with a pure SNN architecture. By substituting spiking neurons with softmax and activating Query, Key, and Value with spiking neurons, it offers a spiking self-attention method that does away with multiplication. Additionally, it substitutes batch normalization and

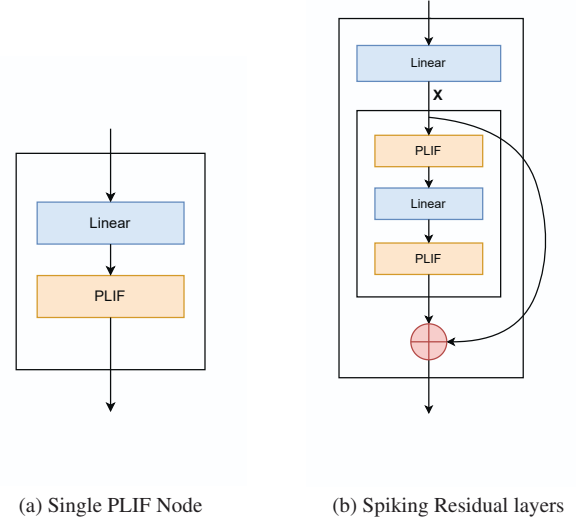


Figure 1. Variants of Spiking Temporal processor

spiking neurons for layer normalization and GELU activation in the Transformer. By altering the residual connection paradigm, Spikingformer [27], based on Spikformer, produces a Vision Transformer solely driven by spikes. A spike-driven self-attention method with linear complexity is proposed by Spike-driven Transformer [26], which successfully lowers energy usage. SpikingResformer [20] combines ResNet architecture with Dual spike self-attention.

Vision Feature extractors: Pretrained models like ResNet [9] have demonstrated remarkable success in image classification by extracting rich spatial features. However, their direct application to event data is nontrivial due to the temporal nature of DVS streams. Combining SNNs with traditional deep learning models has shown promise in enhancing temporal and spatial representations, though few studies fully integrate these paradigms.

3. Preliminary

We describe the Spiking neuron models used in this paper and self-attention framework.

3.1. Spiking neuron models

Numerous models of spiking neurons [25] have been put forth, and they exhibit trade-offs between computational feasibility and biological accuracy. Based on our requirements, we can choose the appropriate neuron.

3.2. Leaky Integrate and Fire (LIF) Model

The integrate-and-fire [25] model integrates the input current over time until the membrane potential exceeds a

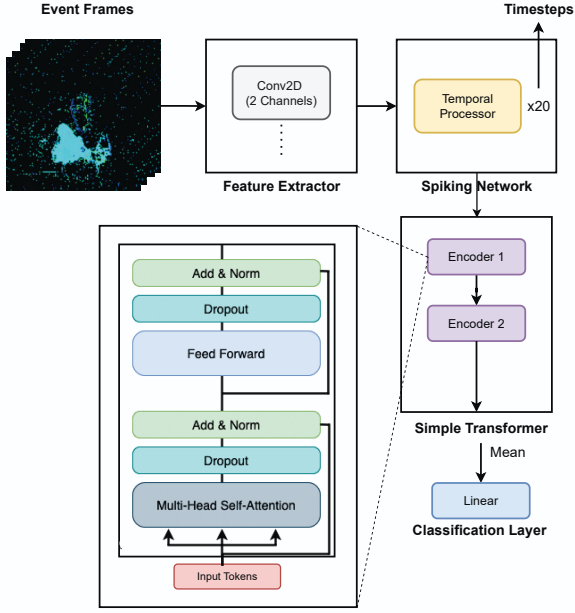


Figure 2. Simple Transformer model architecture.

threshold without accounting for the behaviour of the biological ion channel. By adding a leak factor to this model, the LIF model captures the diffusion of ions through the membrane when a certain equilibrium is not achieved in the cell. The LIF model and its variations are among the most popular applications of the spiking neuron model due to its ease of use and low computational cost. The dynamics of LIF Node [6]

$$\tau \frac{dV(t)}{dt} = -(V(t) - V_{\text{rest}}) + X(t), \quad (1)$$

where:

- $V(t)$ represents the membrane potential at time t .
- V_{rest} is the resting membrane potential.
- τ is the membrane time constant.
- $X(t)$ denotes the external input current at time t .

The membrane potential $V(t)$ returns to a reset value $V_{\text{reset}} < V_{\text{th}}$ after the neuron elicits a spike when it above a specific threshold V_{th} at time t^f .

3.3. Parametric Leaky Integrate-and-Fire (PLIF) Model

The PLIF model [6] extends the LIF framework by introducing a learnable parameter for the membrane time constant. This parameter adapts dynamically during training. The dynamics of the PLIF neuron can be described as:

$$H_t = V_{t-1} + k(a) \left(- (V_{t-1} - V_{\text{reset}}) + X_t \right), \quad (2)$$

where:

- H_t is the membrane potential after neuronal dynamics at timestep t .
- $k(a)$ is a clamping function ensuring numerical stability, typically implemented as a sigmoid:

$$k(a) = \frac{1}{1 + \exp(-a)}.$$

- a is a learnable parameter representing the membrane time constant indirectly.

Analogy to RNNs The LIF and PLIF models can be interpreted as recurrent neural networks (RNNs), where:

- The integration term $\frac{1}{\tau} X_t$ enables the neuron to encode and accumulate information from current inputs.
- The leakage term $(1 - \frac{1}{\tau}) V_{t-1}$ serves as a forgetting mechanism for prior membrane potentials.
- In PLIF, the adaptive τ allows for dynamic control over the balance between memory retention and input integration.

This analogy highlights the similarity between spiking neurons with trainable dynamics and the gating mechanisms in models like Long Short-Term Memory (LSTM) networks, where the gates control the flow of information.

3.4. Multi-head Attention

Multiple heads Attention is a module for attention mechanisms that executes many simultaneous attention processes. The predicted dimension is obtained by concatenating and linearly transforming the separate attention outputs. Multiple attention heads intuitively enable diverse attention to different parts of the sequence (e.g., longer-term dependencies versus shorter-term dependencies).

For multi-head attention, we compute attention across h heads:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) W^O \quad (3)$$

where each attention head is defined as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (4)$$

All W are learnable parameters and

- $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$: Projection matrix for the queries.
- $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$: Projection matrix for the keys.
- $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$: Projection matrix for the values.
- $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$: Projection matrix for the concatenated output.

The input is projected into queries, keys, and values using learned weight matrices W^Q , W^K , and W^V . Each head computes attention independently on the projected inputs. The outputs of all attention heads are concatenated. The concatenated result is projected back into the model’s dimensional space using W^O .

4. Methods

In this section, we define the network architecture and overall pipeline.

4.1. Network architecture

The event-based input is converted to frame-level representation with timesteps T of 20. Time dimension is flattened for providing input to the feature extractor. A ResNet backbone pretrained on ImageNet is used for feature extraction. A 512-dimensional feature vector is extracted from the modified final layer of ResNet-18 [9]. The feature is reshaped for processing temporal information and passed to the temporal processor. The spiking processor outputs temporally processed embeddings for each timestep. The temporal embeddings are given as input to a Simple Transformer having few encoder layers. The output from the Simple Transformer is pooled across time and passed to a single linear layer for classification. The overall pipeline is shown in Fig. 2. Ours is a fairly simple transformer architecture which achieves significant results.

4.2. Pretrained Vision Backbone

We have used ResNet [9] as a feature extractor to process frame-based representations of event data. The original ResNet architecture is modified to accept 2-channel polarity as input, corresponding to positive and negative events, by replacing the first convolutional layer :

$$\text{Conv1} : \mathbb{R}^{2 \times H \times W} \rightarrow \mathbb{R}^{64 \times H' \times W'} \quad (5)$$

The fully connected (FC) layer of ResNet is replaced by an identity mapping to output spatial embeddings. We have used pretrained weights, which reduces the need for large-scale training on event datasets, and ensures high generalization capability.

4.3. Spiking Temporal Processor

The temporal spiking processor models temporal dynamics using a stack of PLIF neurons. Temporal processor has a linear layer, whose input and output features are 512 and 256. PLIF node doesn’t change the input dimensions. We have used 2 variants of Temporal Processor. The first variant has a single PLIF node as shown in Fig. 1a. The second variant has 2 PLIF nodes with a linear layer, and residual connection as shown in Fig. 1b. Residual connections

Methods	Architecture	Venue	Accuracy
Event-based	DVS Gesture [1]	CVPR 2017	94.6
	PointNet [23]	WACV 2019	95.3
	Time-Surfaces [15]	WACV 2019	95.3
	Event RNN [22]	ICLR 2023	97.8
	Event Graph [3]	ICLR 2023	97.8
	TOR Volumes [2]	TPAMI 2022	96.2
	Event-transformer [19]	CVPRW 2022	96.2
	GET-transformer [18]	ICCV 2023	97.9
Spiking NN	SLAYER [21]	NeurIPS 2018	93.6
	Learnable Membrane [6]	ICCV 2021	97.6
	Residual learning [5]	NeurIPS 2021	97.9
	SCRNN [24]	Front. Neuro’20	92.0
	Spikingformer [27]	Arxiv 2023	98.3
	Spikformer [28]	ICLR 2023	98.3
	Spike transformer [26]	NeurIPS 2023	99.3
	SpikingResformer [20]	CVPR 2024	93.4
Ours		-	98.3

Table 1. Results on DVS Gesture

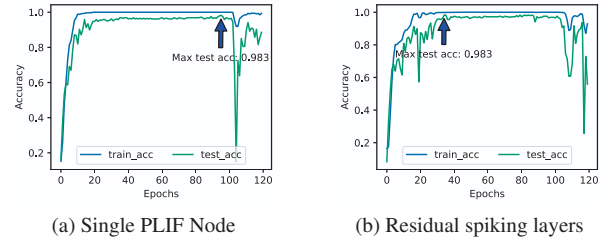


Figure 3. Accuracy plots for variants of temporal processor

between spiking layers ensure gradient flow during training and stabilize training:

$$x^{(l+1)} = \text{PLIF}(Wx^{(l)}) + x^{(l)} \quad (6)$$

4.4. Simple Transformer

To capture long-range temporal dependencies, the simple Transformer employs lightweight encoders with multi-head attention. The simple Transformer has two encoder layers. Detailed architecture of encoders is shown in Fig. 2. Encoder 1 has 4 heads, and Encoder 2 has two sub-encoder layers. The Transformer refines the temporally processed embeddings while maintaining sparse computations. It captures complex temporal relationships, and its multi-head attention mechanism allows the model to focus on important temporal regions.

4.5. Classification Layer

The outputs from the simple Transformer are averaged over the temporal dimension to produce a final embedding:

$$\mathbf{y} = \text{FC} \left(\frac{1}{T} \sum_{t=1}^T y_t \right) \quad (7)$$

Dataset	Methods	Accuracy
N-MNIST	Event Graph [3]	99.0
	Slayer [21]	99.20 $\pm$ 0.02
	Learnable Membrane [6]	99.61
	Data Augment [13]	99.7
	Ours	99.32
CIFAR10-DVS	Event Graph [3]	54.0
	Residual learning [5]	74.4
	Learnable Membrane [6]	74.8
	SpikingResformer [20]	84.8
	Ours	75.9

Table 2. Results on N-MNIST and CIFAR10-DVS Dataset

Method	Parameters (M)	SOPs (G)	Energy (mJ)
Spiking ResNet-34 [10]	21.79	65.28	59.30
Spikformer [28]	66.34	22.09	21.48
Spikingformer [27]	66.34	-	13.68
Spike transformer [26]	66.34	-	6.09
SpikingResformer [20]	60.38	9.74	8.76
Ours	15.32	1.82	8.372

Table 3. Computational Complexity to classify one single image from Imagenet dataset

A final classification layer of fully connected (FC) layer predicts the class probabilities.

5. Experiments

In this section, we evaluate our method on neuromorphic datasets CIFAR10-DVS [12], DVS Gesture [1] and N-MNIST [17]. We also present an ablation study on ResNet feature extractor, LIF, PLIF nodes, different timesteps and residual connections for classification on DVS Gesture.

5.1. Training

For optimization we have used adam, with a learning rate 0.001 and cosine annealing learning rate scheduler with $T_{schedule} = 64$. The batch size is set to 64 for training. For PLIF nodes, we have set $V_{reset} = 0$ and $V_{th} = 1$. We have trained our model on Nvidia A100 GPU for a total of 120 epochs. Single PLIF variant of temporal processor converges in 95 epochs, and Residual variant converges in 34 epochs on DVS gesture dataset. The model architecture is same for all the datasets.

5.2. DVS Gesture

The DVS Gesture [1] dataset contains 11 hand and arm gestures gathered from 29 subjects in 122 trials under three lighting conditions. Each participant executed every motion while standing against a stationary background for each trial. The gestures include arm rotations, air guitar, air

drum, hand waving, forearm rolling and others. Each gesture lasts for 6 seconds. Using spikngjelly [4] library, we obtain 1176 train samples and 288 test samples, totalling 1464 overall. These samples are converted into frames of 16 and 20 timesteps having input resolution of 128x128.

5.3. CIFAR10-DVS

CIFAR10-DVS [12] is a neuromorphic version of popular dataset CIFAR-10. Using a dynamic vision sensor (DVS), 10,000 frame-based photos were transformed into 10,000 event streams, producing an intermediate, challenging event-stream dataset of 10 classes. The training set contains 9000 samples and the test set contains 1000 samples. The pre-processing and input resolution for CIFAR10-DVS are the same as the DVS Gesture.

5.4. N-MNIST

N-MNIST [17] is a neuromorphic version of MNIST digit classification dataset. All of the MNIST digits were transformed into event streams using an asynchronous time-based image sensor (ATIS). Since every MNIST digit is set against a consistent background, there aren't many background noise occurrences. The training set contains 60000 samples, and the test set contains 10000 samples. The pre-processing for N-MNIST is the same as the DVS Gesture, and input resolution is 34x34.

5.5. Computational Complexity

For a fair comparison, we have modified our model to accept a 3-channel RGB image of size 224x224 for the Imagenet dataset and the last layer to classify 1000 classes. Energy consumption is calculated as [26, 28]. Spiking ResNet [10] is an SCNN with ResNet-based architecture. The rest of the methods [20, 26–28] are Spiking Transformers. As evident from Table 3, our model has low parameters and takes less synaptic operations(SOPs) for a single image inference.

5.6. Ablation Study

We have performed an ablation study on important parts of our method. The study is conducted on the DVS Gesture dataset. We have evaluated our methods with and without the LIF node, and as shown in Table 4, the model with the PLIF node performs better. This is because, upon learning, the membrane time constants of PLIF neurons in various layers may vary, more accurately reflecting the heterogeneity of neurons.

We also experimented with three channel inputs: positive events, negative events, and a dummy channel with the original ResNet-18 architecture. RGB images encode absolute intensity values across three channels, whereas event frames encode relative intensity changes across two polarities. Hence, a third channel that effectively introduces noise

Method	Accuracy
No LIF node	97.57
LIF Node	96.18
Single PLIF Node	98.26
ResNet-18 3 Channel input	97.57
ResNet-50	96.88
Custom feature extractor	89.93
Multi-attention heads=8	95.49
sub layers=4	96.18
Timesteps=16	96.53

Table 4. Ablation Results on DVS Gesture

degrades performance. A deeper architecture, ResNet-50, also lacks behind the proposed model. A custom feature extractor trained on event frames doesn't generalize well. We varied the number of heads and sub layers for the Simple Transformer and found the best combination of 4 heads and two sub-layers for maximum accuracy. Timesteps of 20 can better capture the temporal information for processing on the feature extractor.

5.7. Future Work

The model architecture can be extended to larger event-based datasets to assess generalizability. We can also apply it to real-time video streams or other temporal domains, such as speech or biosignals. Leverage self-supervised or semi-supervised learning methods to reduce reliance on large labelled datasets. Integrating advanced regularization techniques to control spike rates and improve model robustness for spiking neurons. We can further integrate hybrid vision-language models, such as CLIP or multimodal transformers, with spiking components to address multimodal event-based tasks.

6. Conclusion

In this work, we proposed the Simple Transformer, a novel architecture that integrates spatial representational capabilities of pre-trained vision backbones with the temporal processing strengths of Spiking Neural Networks (SNNs). By leveraging the pre-trained ResNet backbone for spatial feature extraction, the Spiking Temporal processor for biologically plausible temporal dynamics, and the Simple Transformer for capturing long-range temporal dependencies, the Simple Transformer achieves a unique combination of accuracy and energy efficiency.

Our experiments also highlight the benefits of using spiking layers for sparse and efficient computations. The proposed architecture bridges the gap between large pre-trained models and SNNs, showing how these paradigms can complement each other in event-based vision tasks. The simplicity and modularity of the Simple Transformer make

it easy to implement and adapt to other spatiotemporal datasets.

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References

- [1] Arnon Amir, Brian Taba, David Berg, Timothy Melano, Jeffrey McKinstry, Carmelo Di Nolfo, Tapan Nayak, Alexander Andreopoulos, Guillaume Garreau, Marcela Mendoza, Jeff Kusnitz, Michael Debole, Steve Esser, Tobi Delbruck, Myron Flickner, and Dharmendra Modha. A low power, fully event-based gesture recognition system. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 4, 5
- [2] R. Wes Baldwin, Ruixu Liu, Mohammed Almatrafi, Vijayan Asari, and Keigo Hirakawa. Time-ordered recent event (tore) volumes for event cameras. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023. 1, 2, 4
- [3] Yin Bi, Aaron Chadha, Alhabib Abbas, Eirina Bourtsoulatze, and Yiannis Andreopoulos. Graph-based spatio-temporal feature learning for neuromorphic vision sensing. *IEEE Transactions on Image Processing*, 2020. 1, 2, 4, 5
- [4] Wei Fang, Yanqi Chen, Jianhao Ding, Zhaofei Yu, Timothée Masquelier, Ding Chen, Liwei Huang, Huihui Zhou, Guoqi Li, and Yonghong Tian. Spikingjelly: An open-source machine learning infrastructure platform for spike-based intelligence. *Science Advances*, 9, 2023. 5
- [5] Wei Fang, Zhaofei Yu, Yanqi Chen, Tiejun Huang, Timothée Masquelier, and Yonghong Tian. Deep residual learning in spiking neural networks. In *Advances in Neural Information Processing Systems*, volume 34, 2021. 2, 4, 5
- [6] Wei Fang, Zhaofei Yu, Yanqi Chen, Timothée Masquelier, Tiejun Huang, and Yonghong Tian. Incorporating learnable membrane time constant to enhance learning of spiking neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2661–2671, 2021. 1, 2, 3, 4, 5
- [7] Lang Feng, Qianhui Liu, Huajin Tang, De Ma, and Gang Pan. Multi-level firing with spiking ds-resnet: Enabling better and deeper directly-trained spiking neural networks. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI-22*, pages 2471–2477, 2022. 2
- [8] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, and Davide Scaramuzza. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022. 1

- [9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 2, 4
- [10] Yangfan Hu, Huajin Tang, and Gang Pan. Spiking deep residual networks. *IEEE Transactions on Neural Networks and Learning Systems*, 34, 2023. 2, 5
- [11] Nikola K Kasabov. *Time-space, spiking neural networks and brain-inspired artificial intelligence*. Springer, 2019. 1
- [12] Hongmin Li, Hanchao Liu, Xiangyang Ji, Guoqi Li, and Luping Shi. Cifar10-dvs: An event-stream dataset for object classification. *Frontiers in Neuroscience*, 11, 2017. 5
- [13] Yuhang Li, Youngeun Kim, Hyoungseob Park, Tamar Geller, and Priyadarshini Panda. Neuromorphic data augmentation for training spiking neural networks. In *European Conference on Computer Vision*, pages 631–649. Springer, 2022. 2, 5
- [14] Wolfgang Maass. Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, 1997. 1
- [15] Jean-Matthieu Maro, Sio-Hoi Ieng, and Ryad Benosman. Event-based gesture recognition with dynamic background suppression using smartphone computational capabilities. *Frontiers in Neuroscience*, 2020. 4
- [16] Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36, 2019. 2
- [17] Garrick Orchard, Ajinkya Jayawant, Gregory K. Cohen, and Nitish Thakor. Converting static image datasets to spiking neuromorphic datasets using saccades. *Frontiers in Neuroscience*, 9, 2015. 5
- [18] Yansong Peng, Yueyi Zhang, Zhiwei Xiong, Xiaoyan Sun, and Feng Wu. Get: Group event transformer for event-based vision. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 1, 2, 4
- [19] Alberto Sabater, Luis Montesano, and Ana C. Murillo. Event transformer. a sparse-aware solution for efficient event data processing. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022. 1, 2, 4
- [20] Xinyu Shi, Zecheng Hao, and Zhaoifei Yu. Spikingresformer: Bridging resnet and vision transformer in spiking neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 2, 4, 5
- [21] Sumit B Shrestha and Garrick Orchard. Slayer: Spike layer error reassignment in time. *Advances in neural information processing systems*, 31, 2018. 1, 4, 5
- [22] Anand Subramoney, Khaleelulla Khan Nazeer, Mark Schöne, Christian Mayr, and David Kappel. Efficient recurrent architectures through activity sparsity and sparse back-propagation through time. In *The Eleventh International Conference on Learning Representations*, 2023. 1, 2, 4
- [23] Qinyi Wang, Yexin Zhang, Junsong Yuan, and Yilong Lu. Space-time event clouds for gesture recognition: From rgb cameras to event cameras. In *2019 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019. 2, 4
- [24] Yannan Xing, Gaetano Di Caterina, and John Soraghan. A new spiking convolutional recurrent neural network (scrnn) with applications to event-based hand gesture recognition. *Frontiers in Neuroscience*, 14, 2020. 1, 2, 4
- [25] Kashu Yamazaki, Viet-Khoa Vo-Ho, Darshan Bulsara, and Ngan Le. Spiking neural networks and their applications: A review. *Brain Sciences*, 2022. 1, 2
- [26] Man Yao, Jiakui Hu, Zhaokun Zhou, Li Yuan, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, 2024. 2, 4, 5
- [27] Chenlin Zhou, Liutao Yu, Zhaokun Zhou, Zhengyu Ma, Han Zhang, Huihui Zhou, and Yonghong Tian. Spikingformer: Spike-driven residual learning for transformer-based spiking neural network, 2023. 2, 4, 5
- [28] Zhaokun Zhou, Yuesheng Zhu, Chao He, Yaowei Wang, Shuicheng YAN, Yonghong Tian, and Li Yuan. Spikformer: When spiking neural network meets transformer. In *The Eleventh International Conference on Learning Representations*, 2023. 2, 4, 5